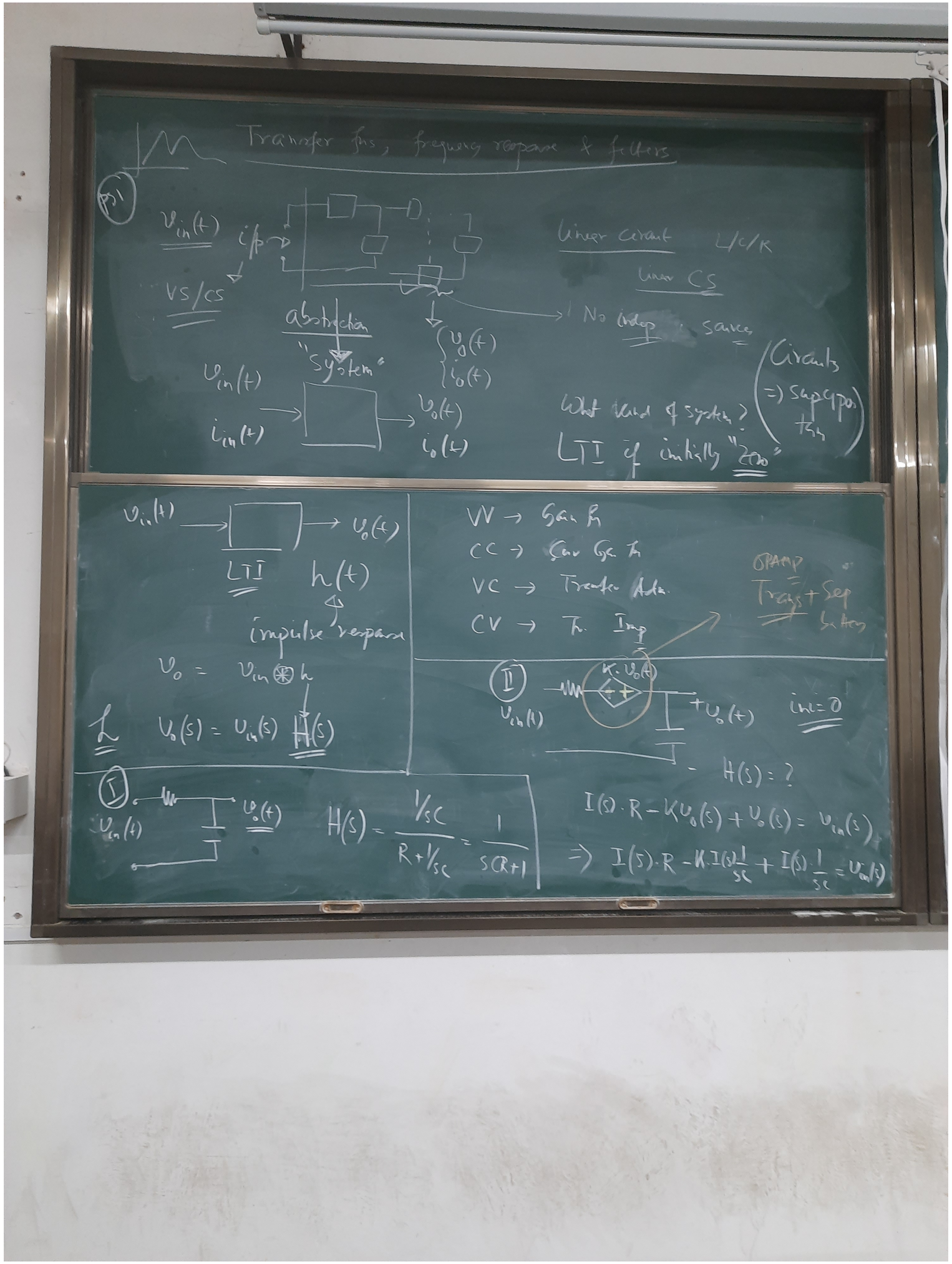


Transfer fns, frequency response & filters

ps1

$v_{in}(t)$ i/p → VS/CS

abstraction

"System"

$\left\{ \begin{array}{l} v_o(t) \\ i_o(t) \end{array} \right.$

$v_{in}(t)$
$i_{in}(t)$ → ☐ → $v_o(t)$
$i_o(t)$

Linear circuit L/C/R

Linear CS

No indep. source

What kind of system?

LTI if initially "zero"

(Circuits ⇒ Superpos. thm)

$v_{in}(t) \rightarrow$ ☐ $\rightarrow v_o(t)$

LTI $h(t)$

impulse response

$v_o = v_{in} \circledast h$

$\mathcal{L}$ $V_o(s) = V_{in}(s)\,H(s)$

VV → Gain fn

CC → Curr Gain fn

VC → Transfer Adm.

CV → Tr. Imp

OPAMP = Trans + Sep Systems

Ⓘ $v_{in}(t)$ $v_o(t)$

$$H(s) = \frac{1/sC}{R + 1/sC} = \frac{1}{sCR+1}$$

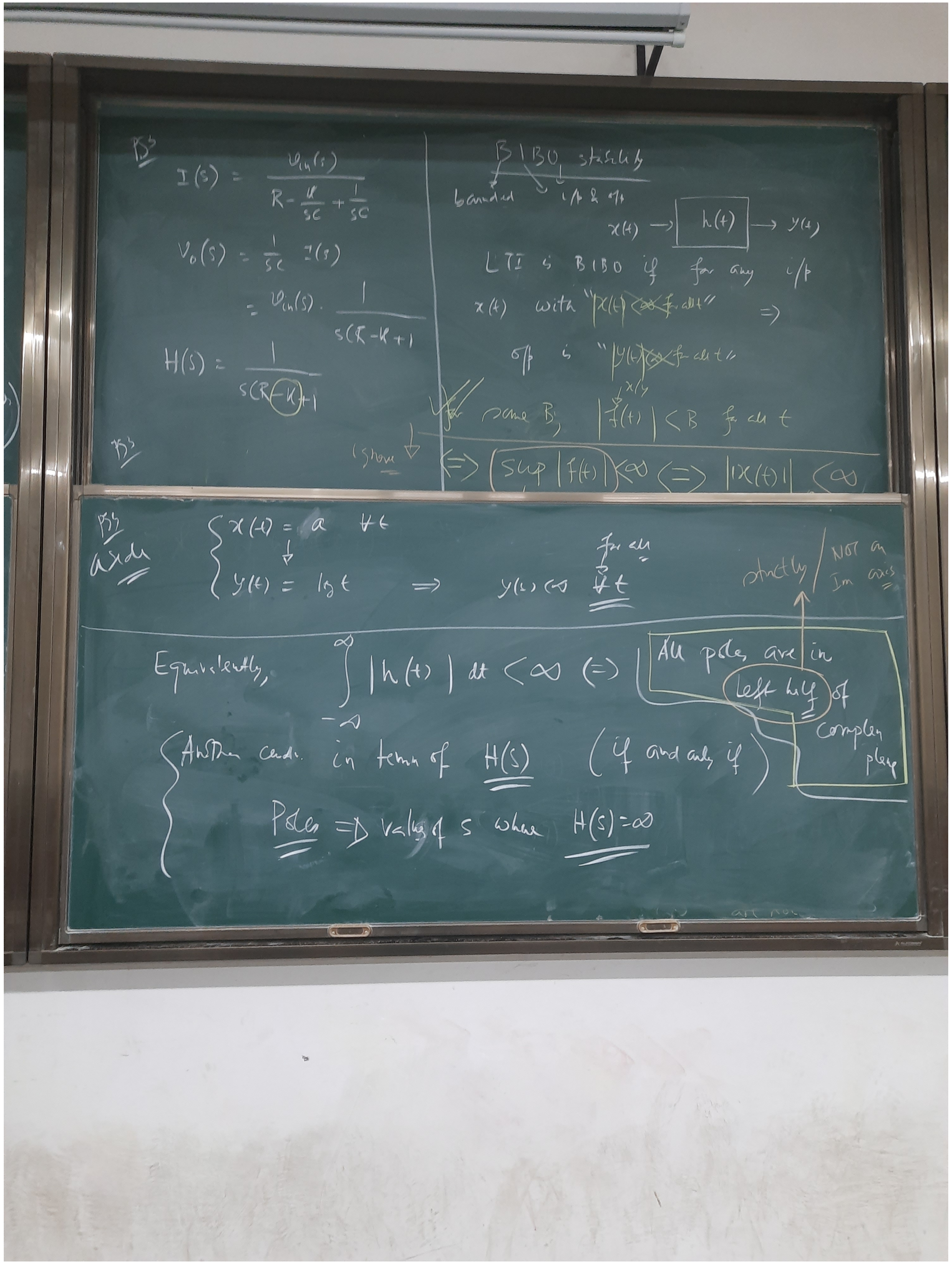
Ⓘ $K \cdot v_o(t)$ $v_{in}(t)$ $+ v_o(t)$ $-$

$i_{in} = 0$

$H(s) = ?$

$$I(s) \cdot R - K V_o(s) + V_o(s) = V_{in}(s)$$

$$\Rightarrow I(s) \cdot R - K \cdot I(s) \frac{1}{sC} + I(s) \frac{1}{sC} = V_{in}(s)$$



$$I(s) = \frac{V_{in}(s)}{R - \frac{K}{sC} + \frac{1}{sC}}$$

$$V_o(s) = \frac{1}{sC} I(s)$$

$$= V_{in}(s) \cdot \frac{1}{s(R - K + 1}$$

$$H(s) = \frac{1}{s(R\boxed{-K}+1}$$

(show ↓

BIBO stability

B → bounded, I/O → i/p & o/p

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$

LTI is BIBO if for any i/p $x(t)$ with "$|x(t)|$ ~~$<\infty$~~ for all t" ⇒

o/p is "$|y(t)|$ ~~$<\infty$~~ for all t"

~~if~~ for some B, $|\overset{x/y}{f}(t)| < B$ for all t

$$\iff \boxed{\sup |f(t)|} < \infty \iff \|x(t)\| < \infty$$

aside

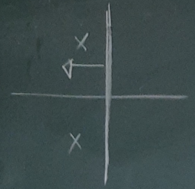
$$\begin{cases} x(t) = a & \forall t \\ \downarrow & \\ y(t) = \log t & \Longrightarrow \quad y(t) < \infty \ \overset{\text{for all}}{\forall} t \end{cases}$$

Equivalently, $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty \iff$ All poles are in left half of complex plane

strictly / NOT on Im axis

Another cond. in terms of $H(s)$ (if and only if)

Poles ⇒ values of s where $H(s) = \infty$



s - complex

H(s)

$\sum_i \frac{A_i}{s-s_i}$ ← Roots of den ⇕ poles

$\mathcal{L}^{-1}$ ⇓

$\sum A_i e^{s_i t}$

① always stable pole = $-1/RC$ in LH

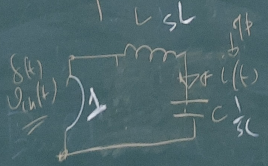
② $K < 1 \Rightarrow$ stable $\boxed{-\frac{1-K}{RC}}$ pole

$K > 1 \Rightarrow$ unstable

If you have passive circuit & a bit of "a resistance" ⇓ stable

$e^{-\frac{R}{2L}t}$, $e^{-\frac{t}{2RC}}$

purely pass unstable



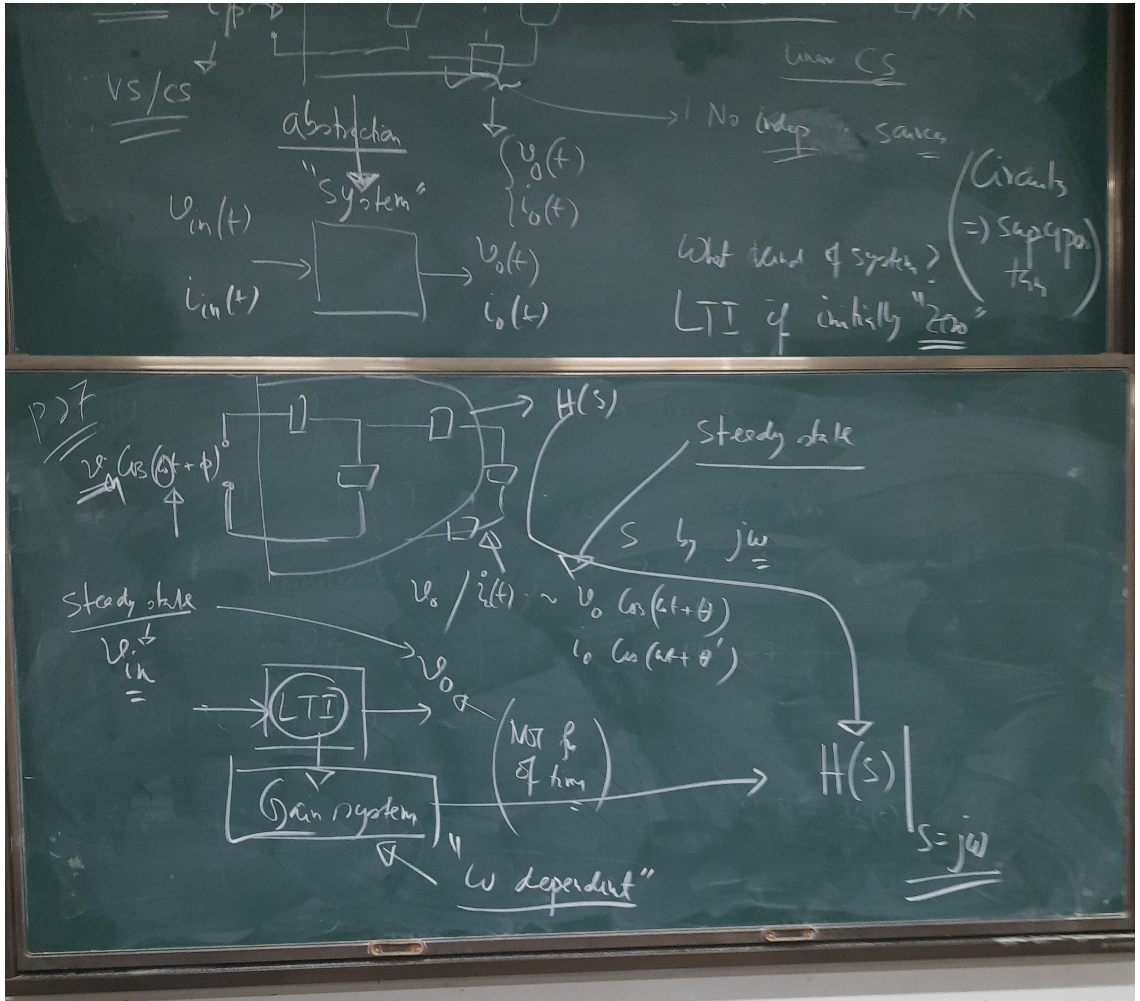
Write Laplace

$i(t) =$

$I(s) = \frac{1}{sL + 1/sC} = K_1 \frac{s}{s^2 + \frac{1}{LC}} = K_1 \cos\left(\frac{1}{\sqrt{LC}}t\right)$

↓

poles on Im axis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linear CS
No indep sources
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p27
$v_i \cos(\omega t+\phi)$
H(s)
Steady state
s by jω
v_o / $i_o(t)$ ~ $v_o \cos(\omega t+\theta)$
$i_o \cos(\omega t+\theta')$
Steady state
v_{in}
v_o
LTI
Gain system
NOT fn of time
"ω dependent"
$H(s)\big|_{s=j\omega}$